NAME: ____________________ **PERIOD:** ________

Reminders and Practice Completion

AP Statistics · Unit 3 · Topic 3.11

[STAR] THE PROBLEM

In a district experiment, 360 volunteers are randomly assigned: 180 get daily reminders and 180 use a standard platform. By Friday, 126 reminder and 105 standard students finish optional practice. For students like these volunteers, a 95% interval for $p_R - p_S$ is $(0.018, 0.215)$.

Iris: "The positive interval and random assignment support a causal increase of 1.8 to 21.5 points for students like these volunteers."

Cole: "There is a 95% chance reminders raise completion for all district students by at least 10 points."

Experiment counts

Version	Complete	Other	Total
Reminder	126	54	180
Standard	105	75	180

FIRST TAKE — one sentence before you work the parts

Which report would you share with classmates? Give one reason from the study.

[BOOK] MATCH EVERY CLAIM TO THE INTERVAL AND DESIGN (keep on the page)

A confidence interval for $p_1 - p_2$ gives plausible values for that population difference in the stated order. If 0 is inside, no difference is plausible; if the whole interval is positive, the evidence supports $p_1 > p_2$. A claimed minimum effect d is supported only when every interval value is at least d. The confidence level is the long-run percent of intervals from repeated use of the method that capture the parameter, not the probability that a fixed parameter or one computed interval is correct. Random assignment can support causation for the experimental units; random sampling is what supports generalization to the sampled population.

Work (a)–(c).

DOK 1 **(a)** State what $p_R - p_S$ represents, and identify whether 0 and 0.10 are contained in the reported interval.

DOK 2 **(b)** Interpret the 95% confidence interval in context. Then interpret the 95% confidence level as the long-run performance of the interval procedure.

DOK 3 **(c)** Choose the warranted conclusion. Use the positions of 0 and 0.10, distinguish random assignment from volunteer recruitment, and diagnose Cole's probability, minimum-effect, and population claims.

> **Frame:** The interval supports a positive effect because ______________________, but it does not establish a 10-point minimum because ________________________.

> **Frame:** Random assignment permits __________________, whereas volunteer recruitment limits ____________________________.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.